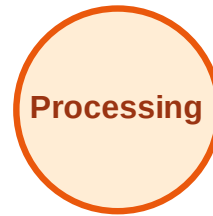


BFS Traversal

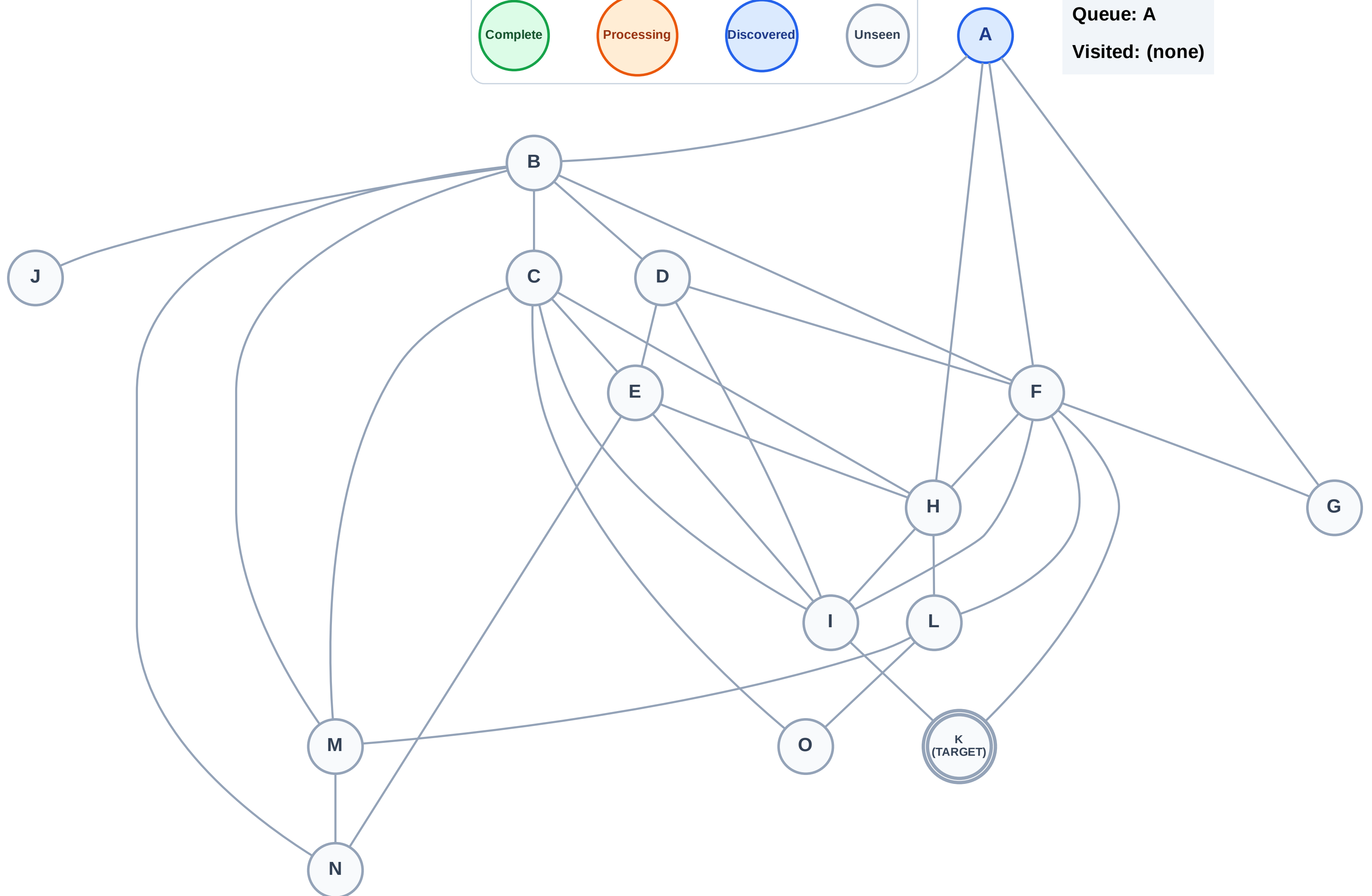
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Queue: A

Visited: (none)



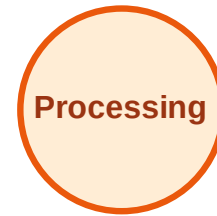
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



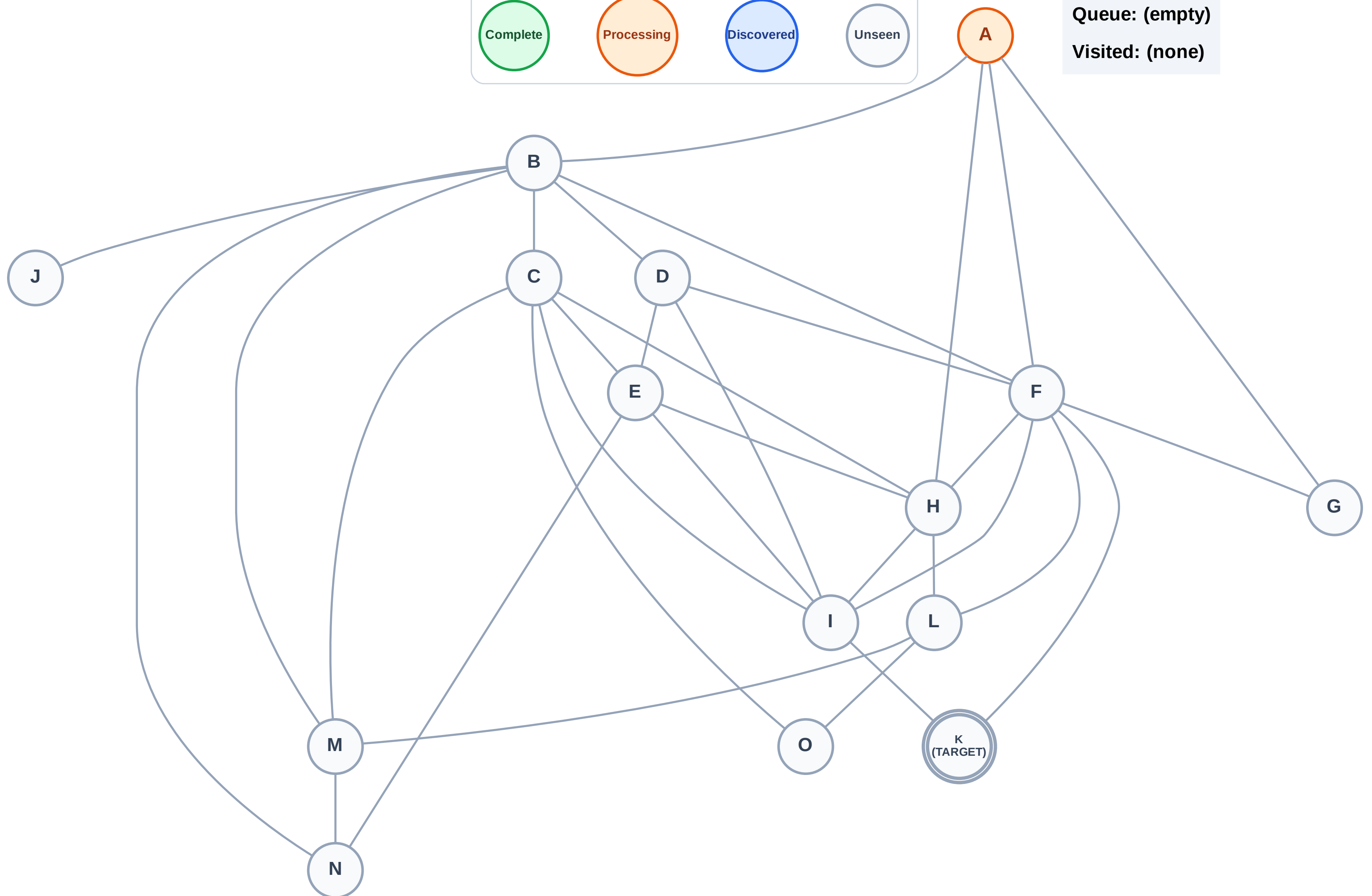
Discovered



Unseen

Queue: (empty)

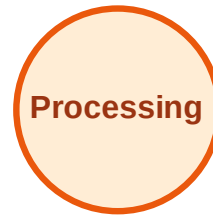
Visited: (none)



BFS Traversal

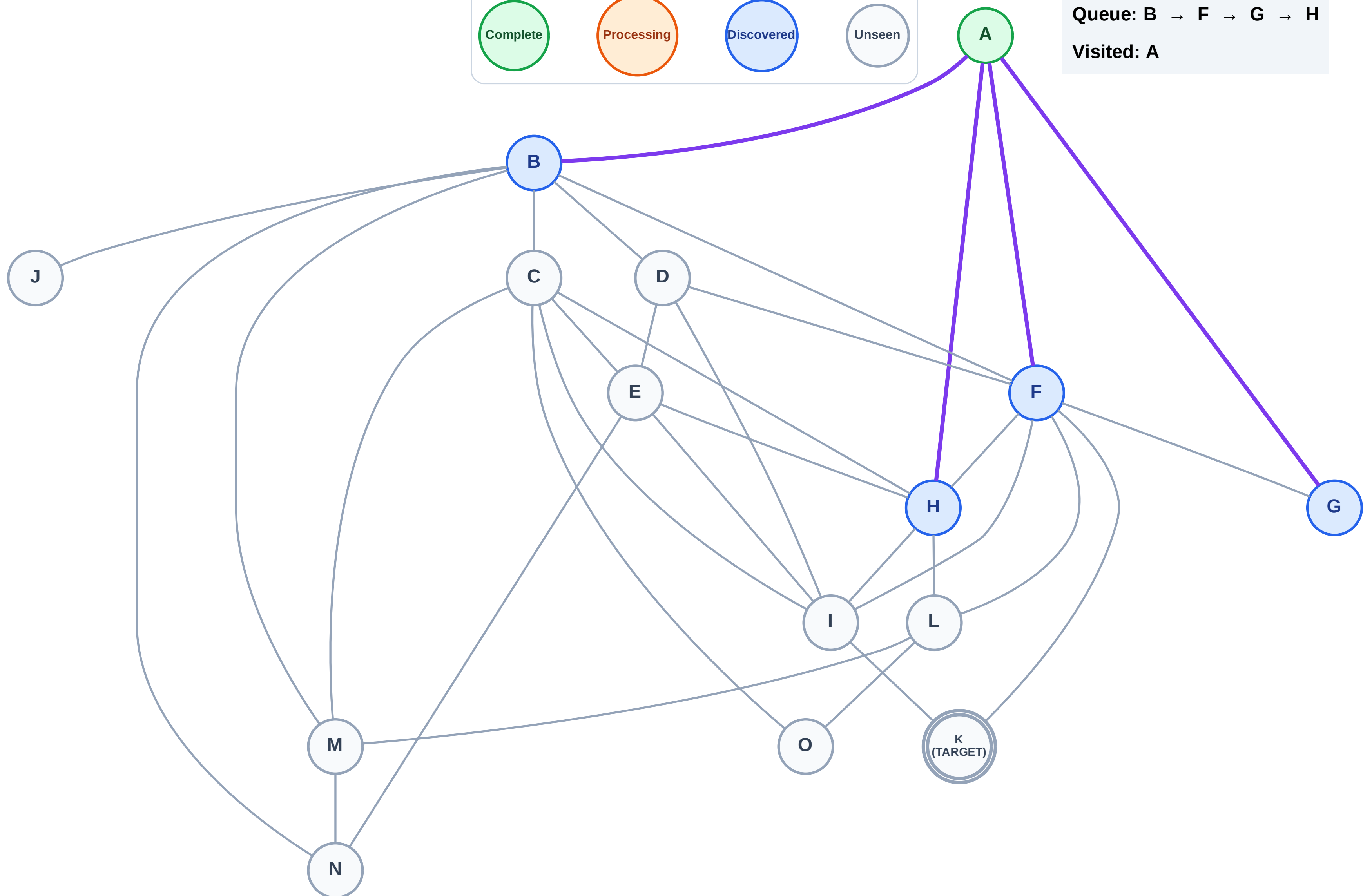
Step 3: Discover B, F, G, H from A

Legend



Queue: B → F → G → H

Visited: A



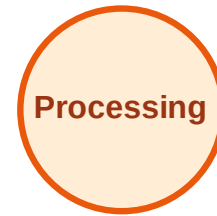
BFS Traversal

Step 4: Process node B

Legend



Complete



Processing



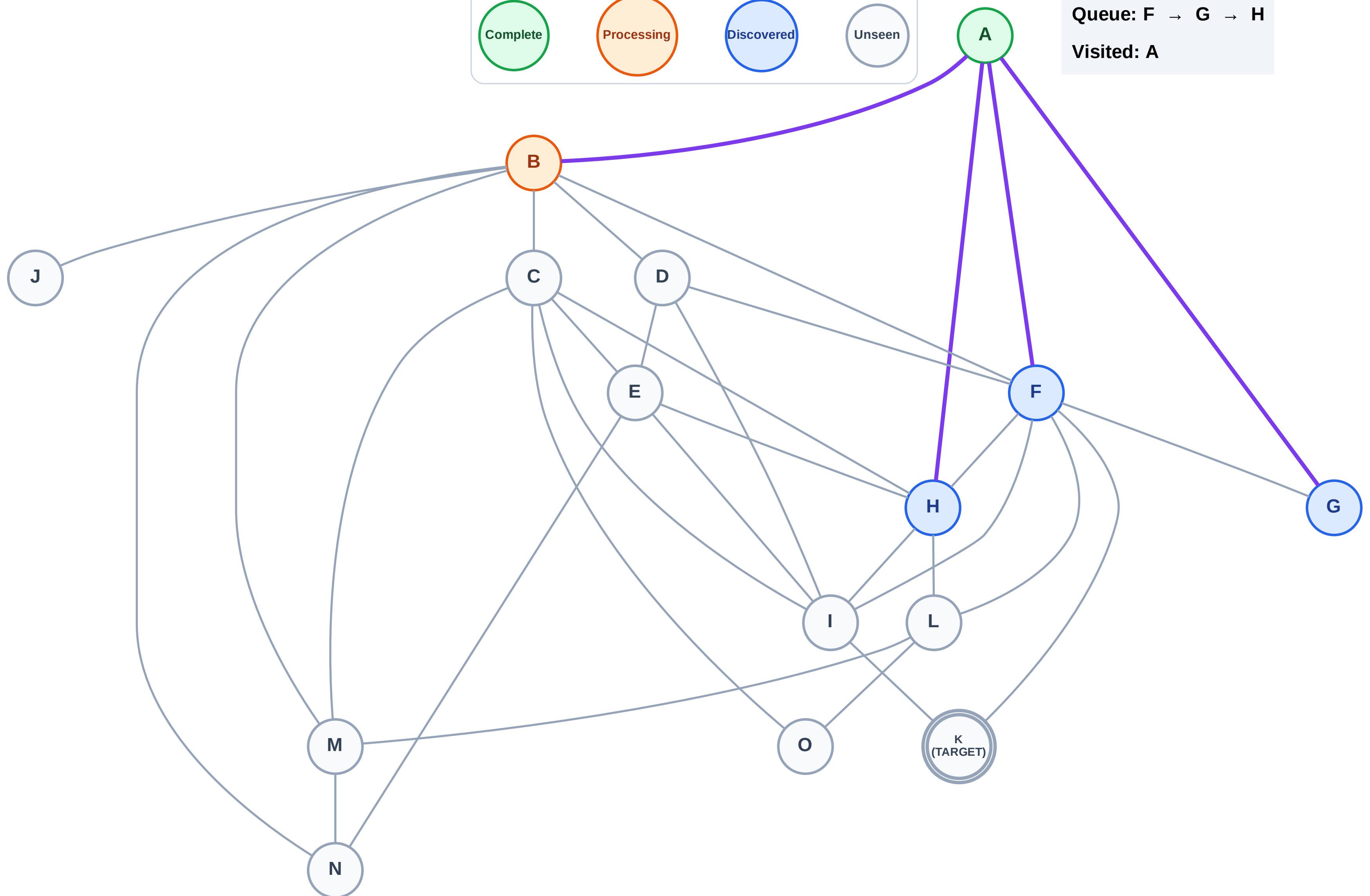
Discovered



Unseen

Queue: F → G → H

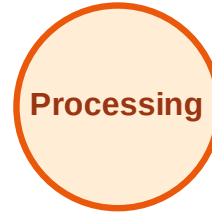
Visited: A



BFS Traversal

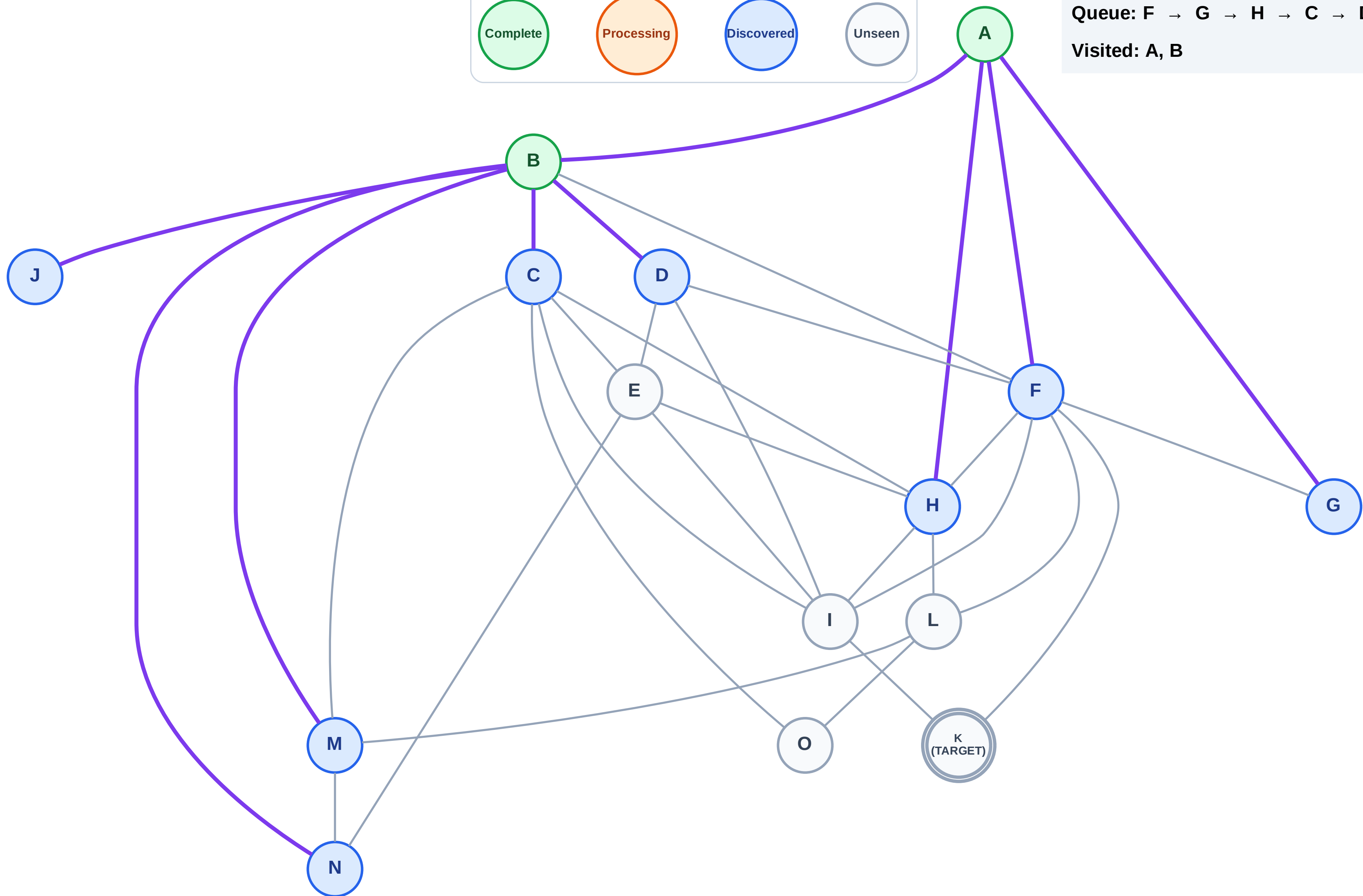
Step 5: Discover C, D, J, M, N from B

Legend



Queue: F → G → H → C → D → J → M → N

Visited: A, B



BFS Traversal

Step 6: Process node F

Legend

Complete

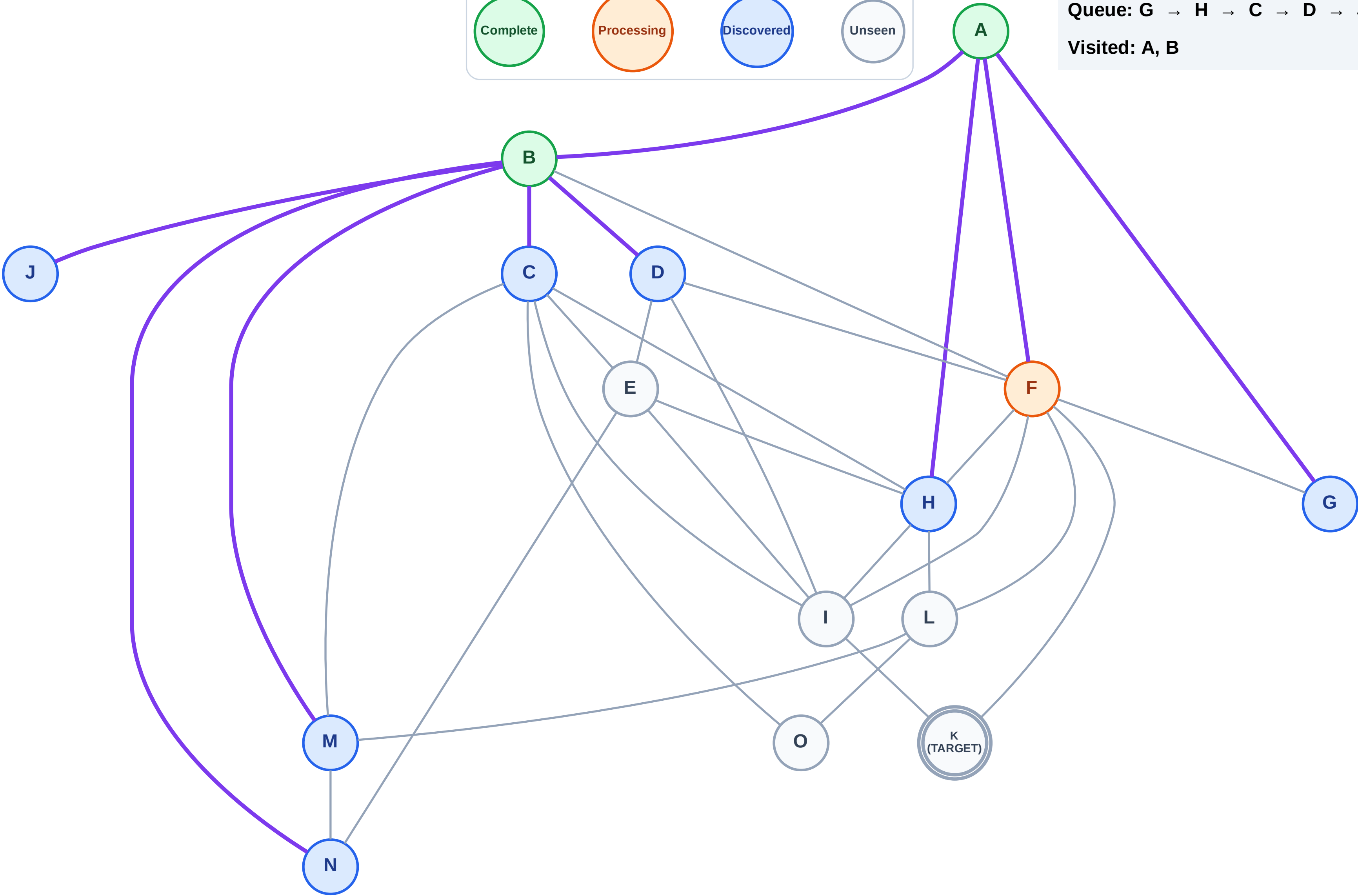
Processing

Discovered

Unseen

Queue: G → H → C → D → J → M → N

Visited: A, B



BFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → F → K

Legend

Complete

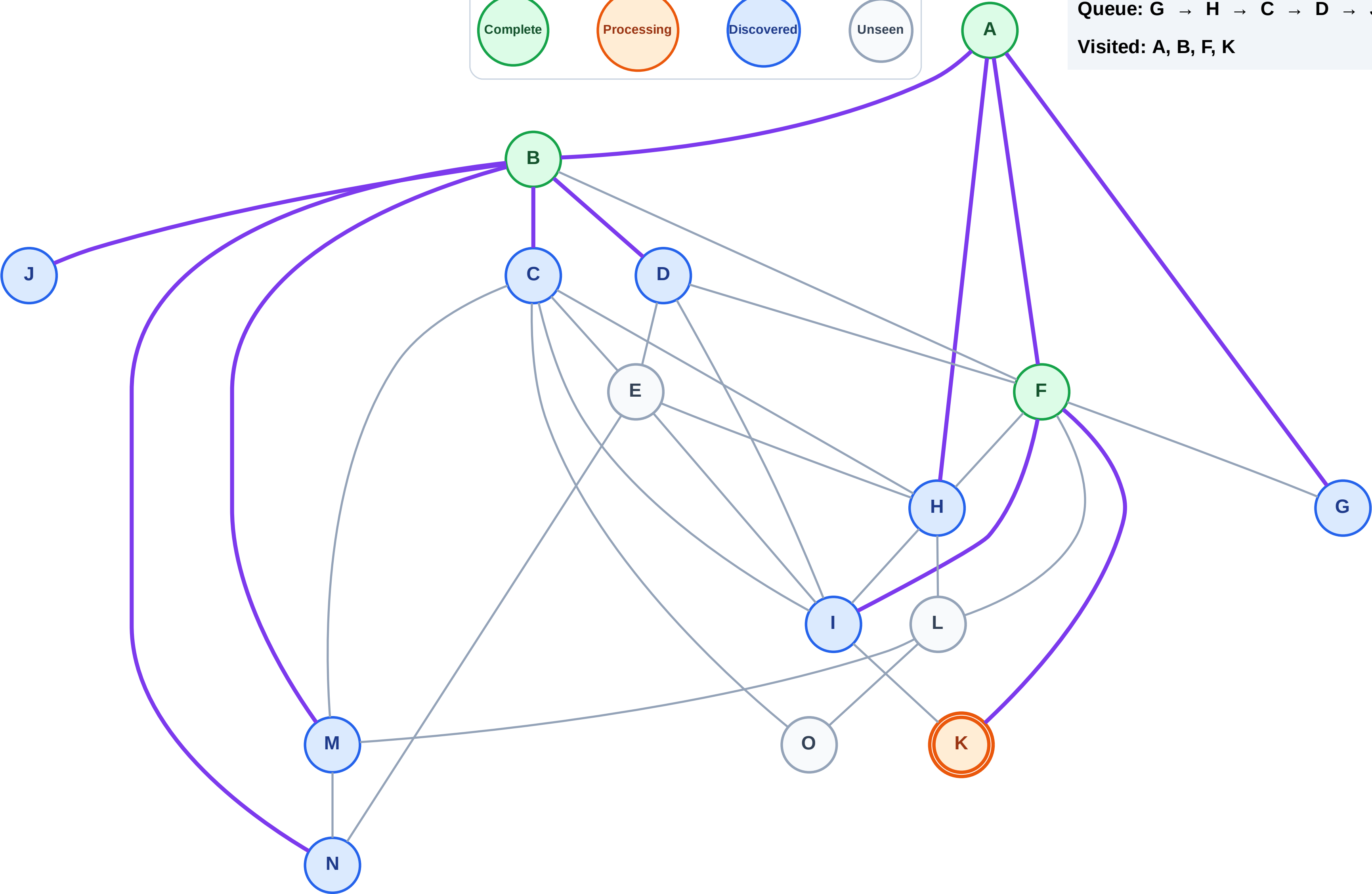
Processing

Discovered

Unseen

Queue: G → H → C → D → J → M → N → I

Visited: A, B, F, K



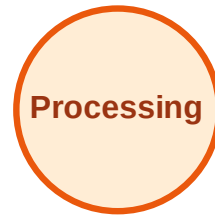
DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Complete



Processing



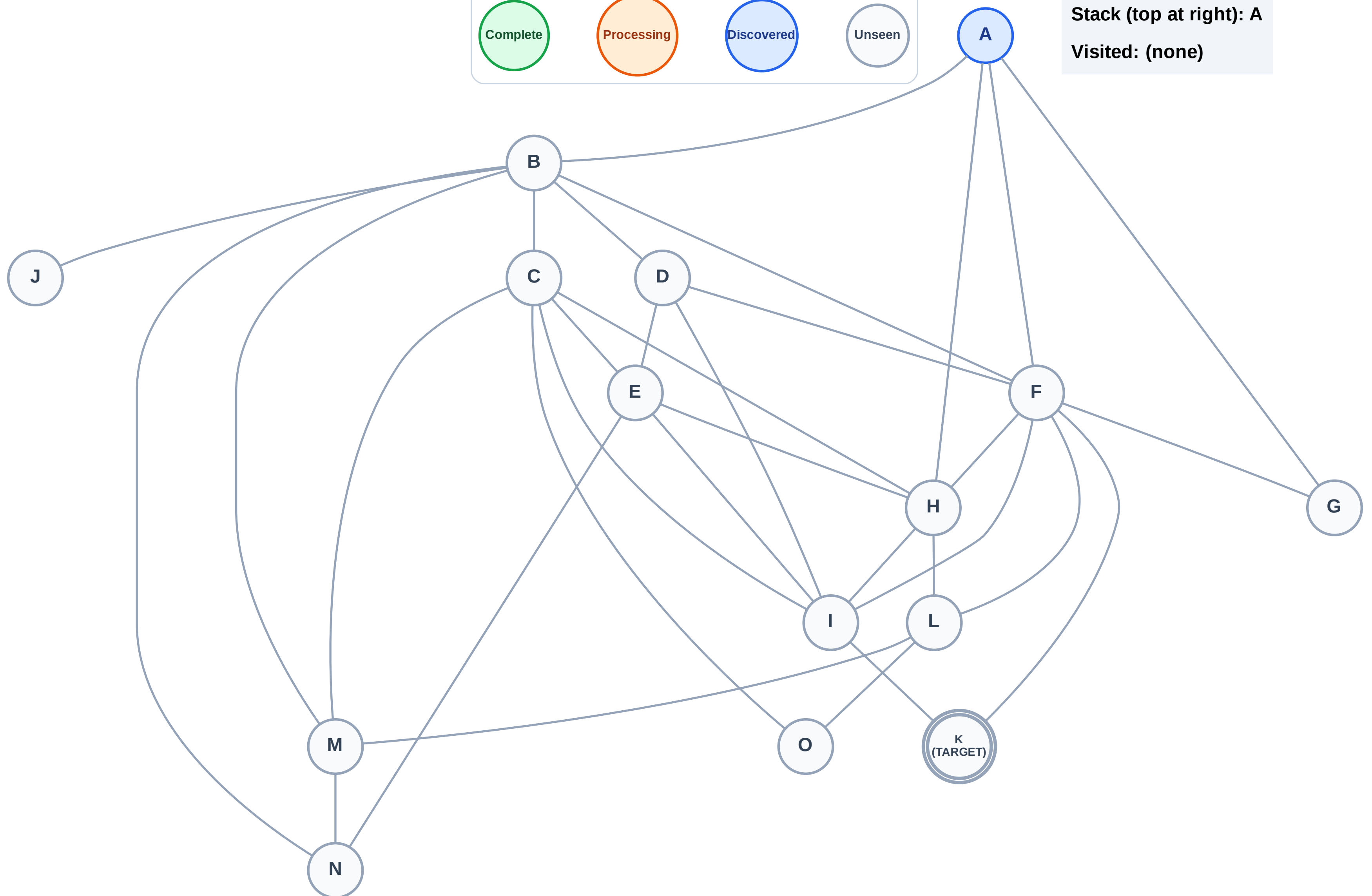
Discovered



Unseen

Stack (top at right): A

Visited: (none)



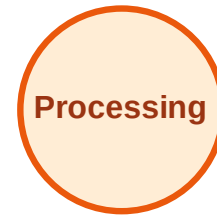
DFS Traversal

Step 2: Process node A

Legend



Complete



Processing



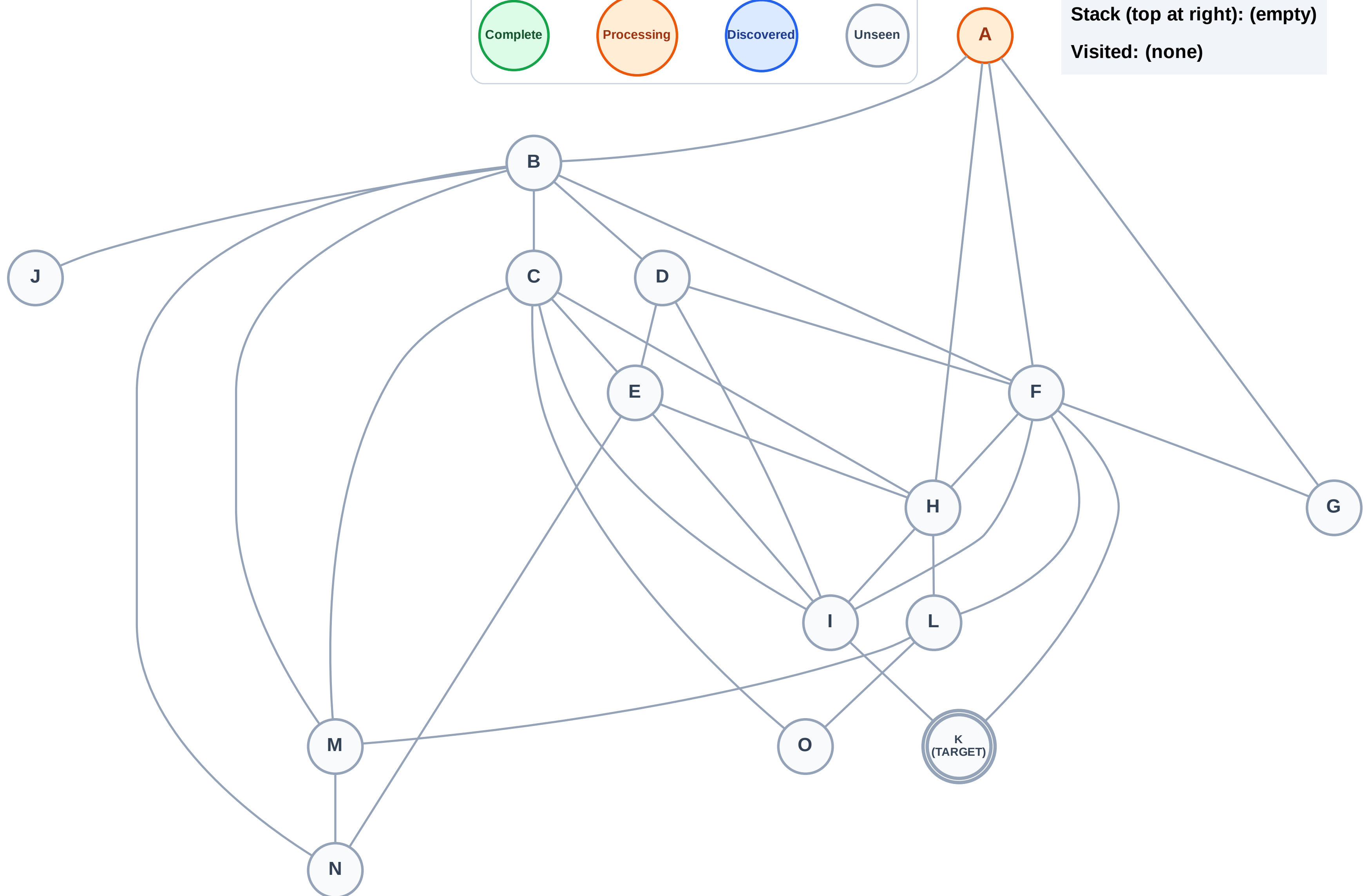
Discovered



Unseen

Stack (top at right): (empty)

Visited: (none)



DFS Traversal

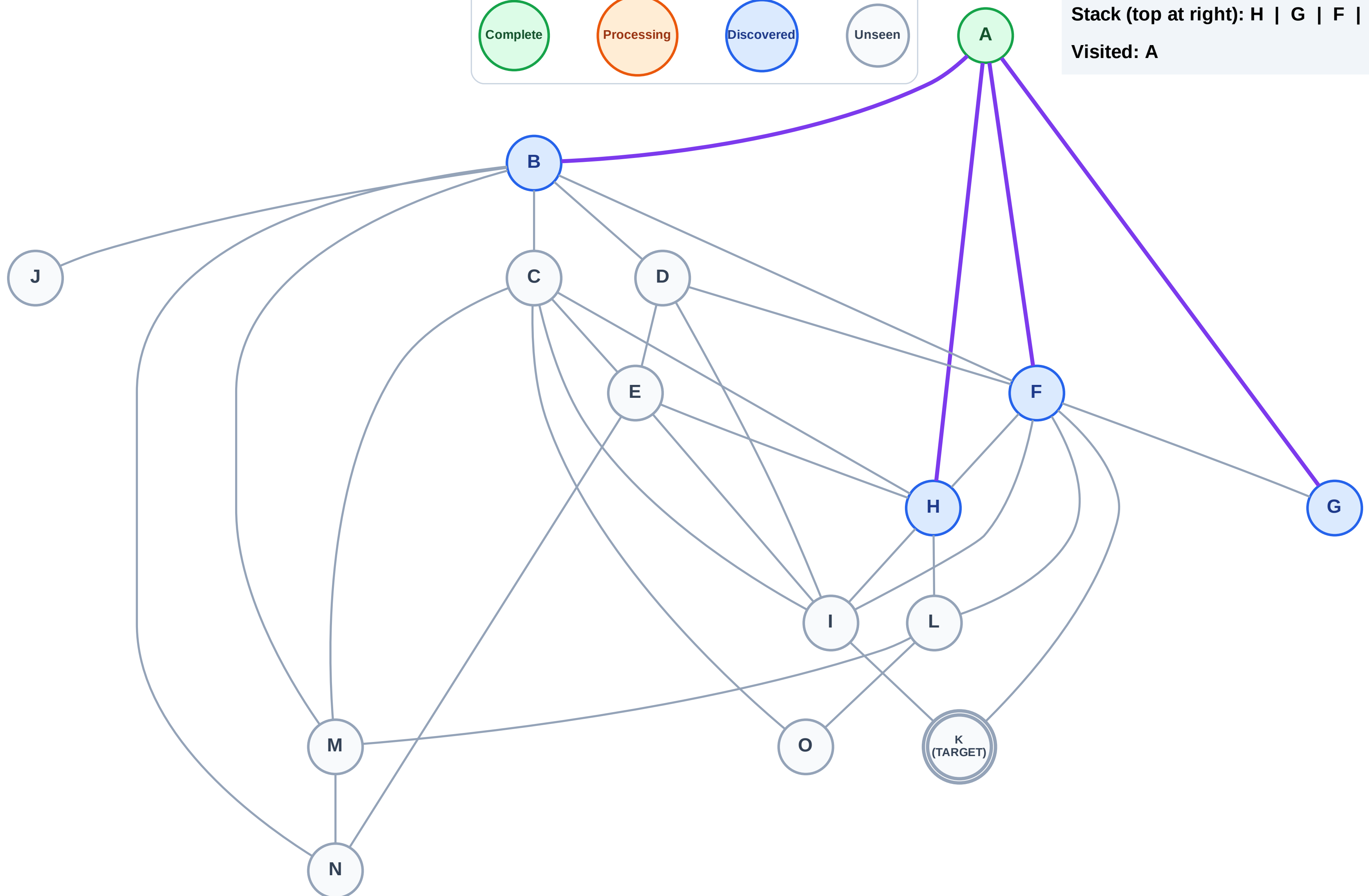
Step 3: Discover H, G, F, B from A

Legend



Stack (top at right): H | G | F | B

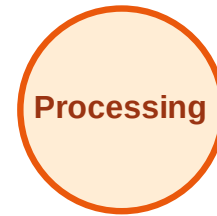
Visited: A



DFS Traversal

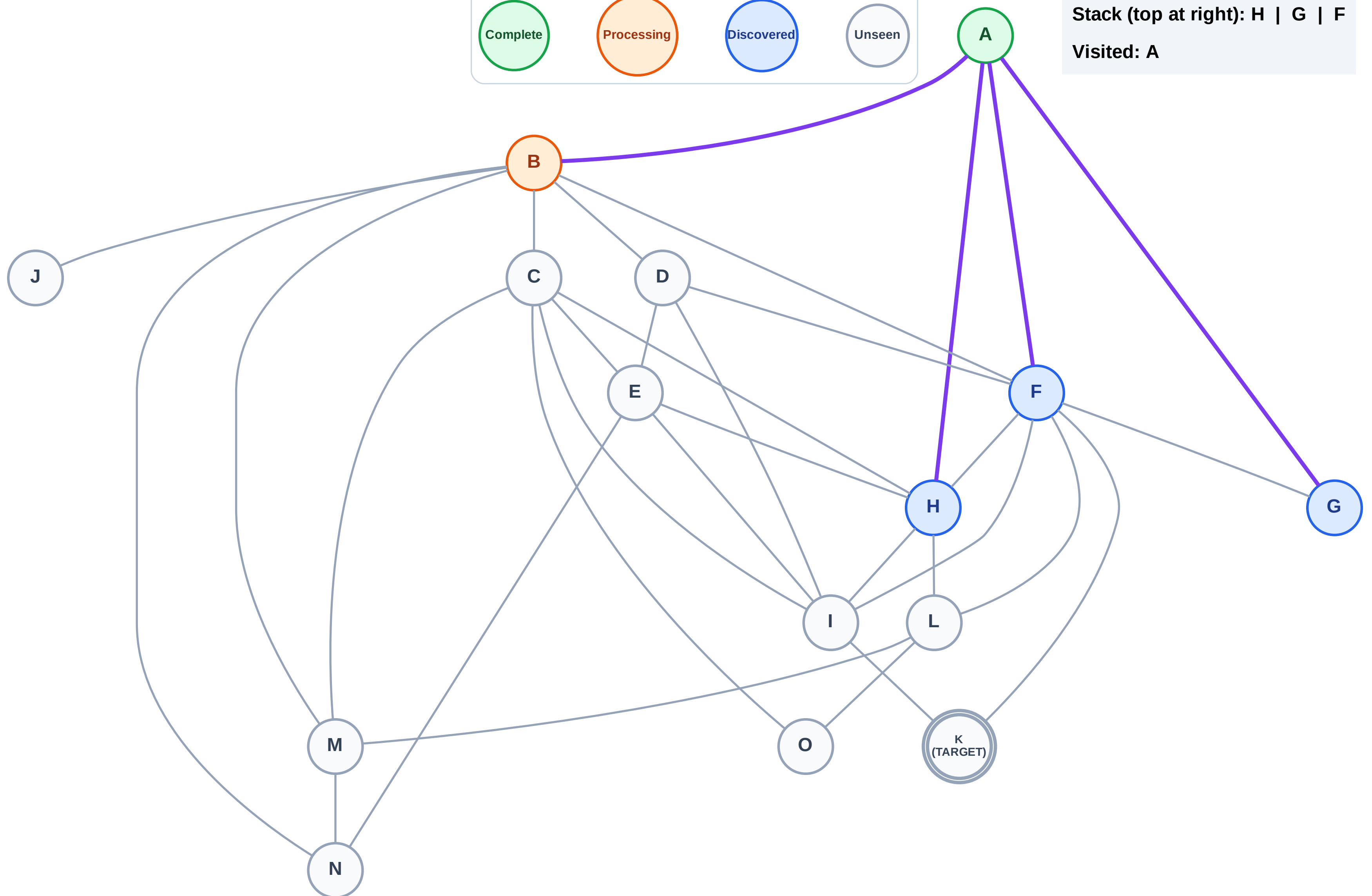
Step 4: Process node B

Legend



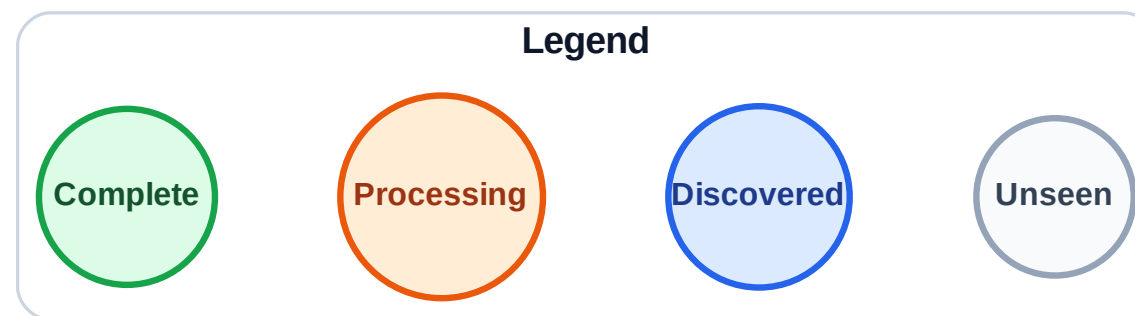
Stack (top at right): H | G | F

Visited: A



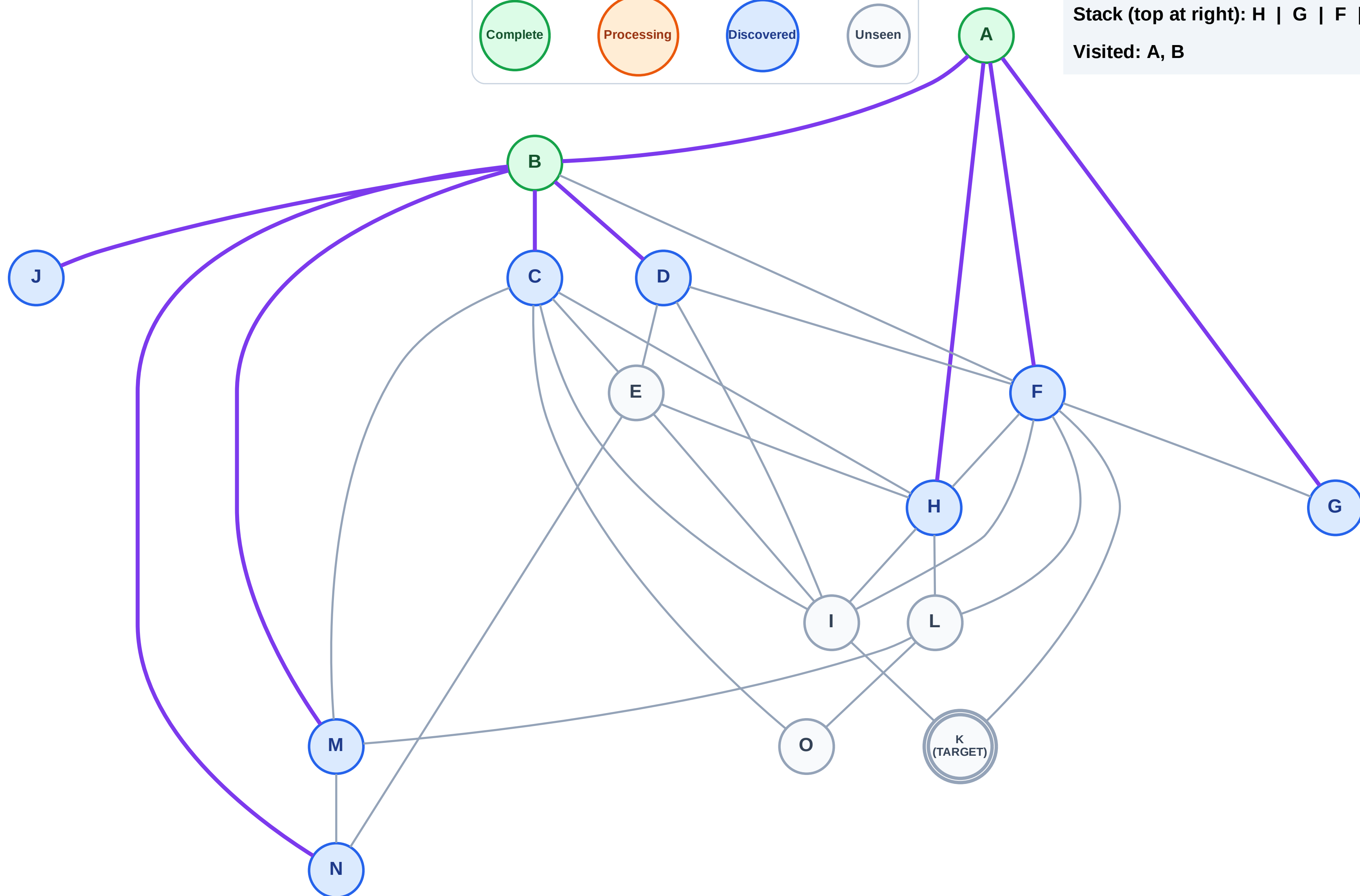
DFS Traversal

Step 5: Discover N, M, J, D, C from B



Stack (top at right): H | G | F | N | M | J | D | C

Visited: A, B



Step 6: Process node C

Step 6: Process node C

Legend

Complete

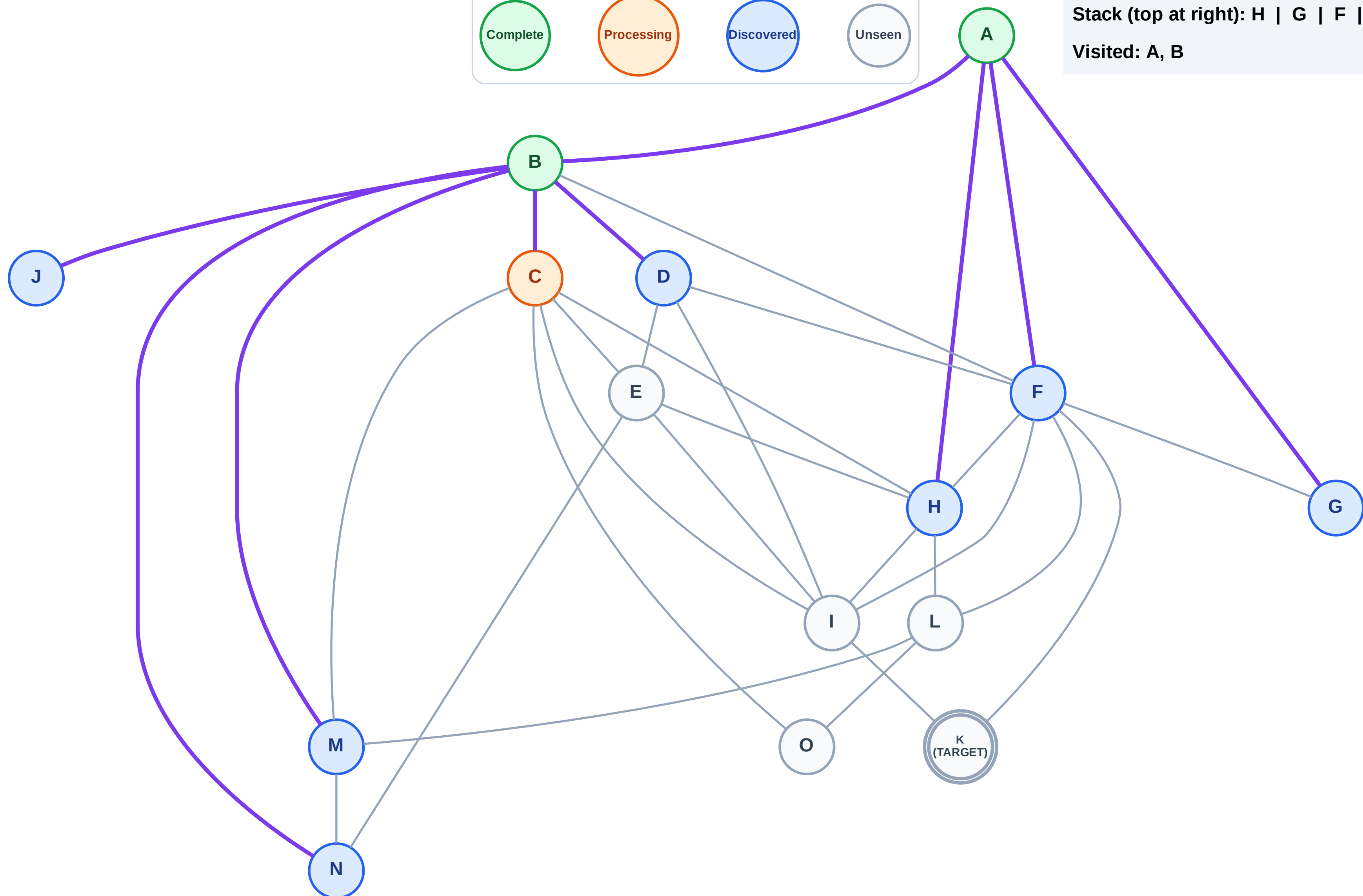
Processing

Discovered

Unseen

Stack (top at right): H | G | F | N | M | J | D

Visited: A, B



DFS Traversal

Step 7: Discover O, I, E from C

Legend

Complete

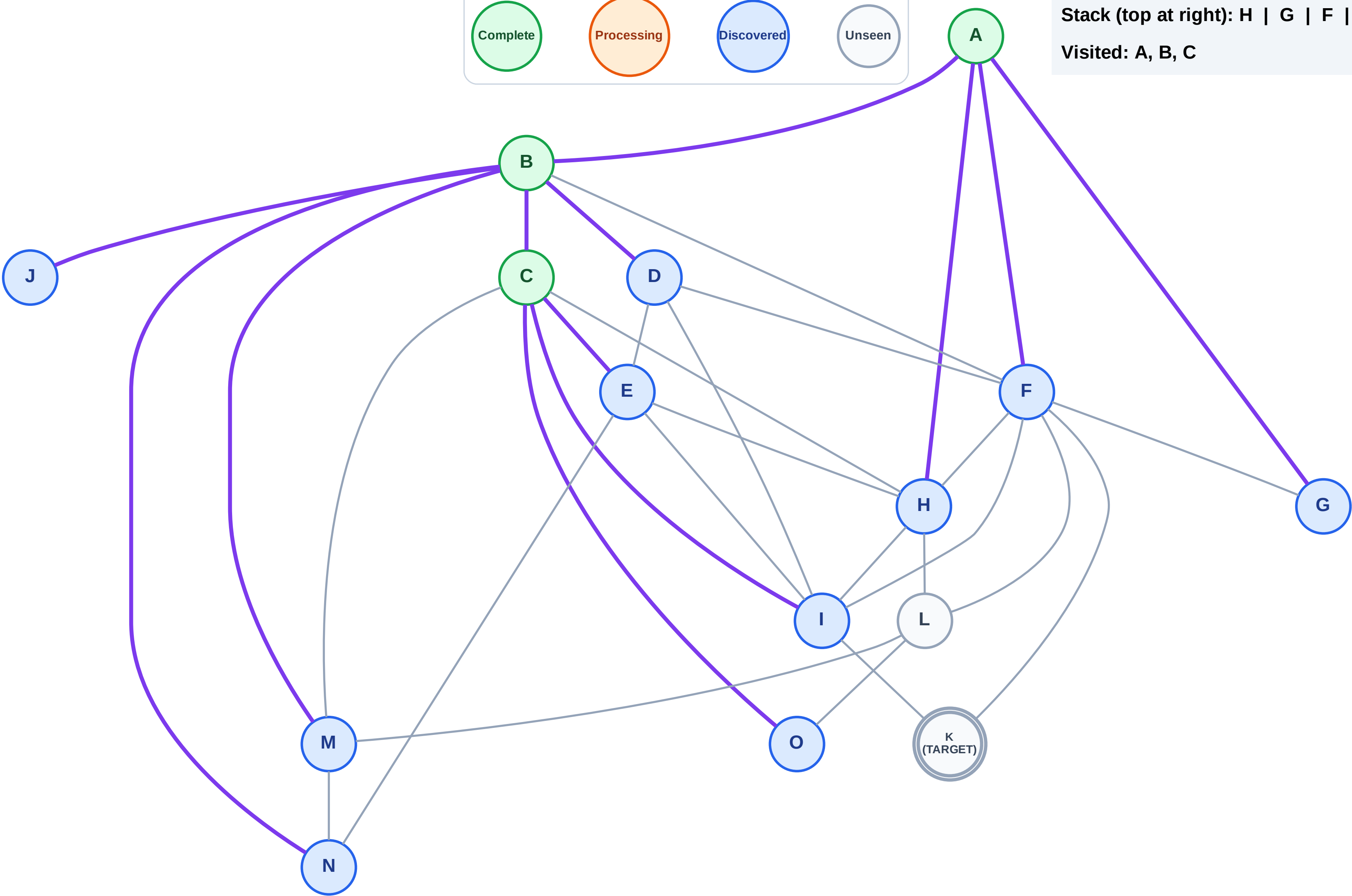
Processing

Discovered

Unseen

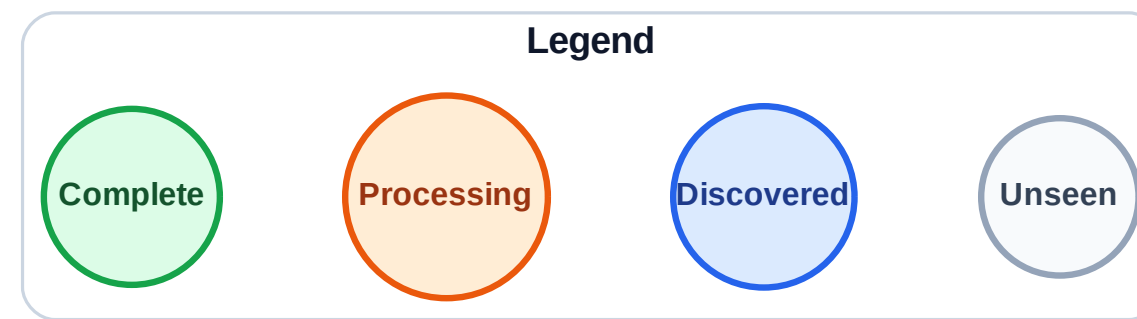
Stack (top at right): H | G | F | N | M | J | D | O | I | E

Visited: A, B, C



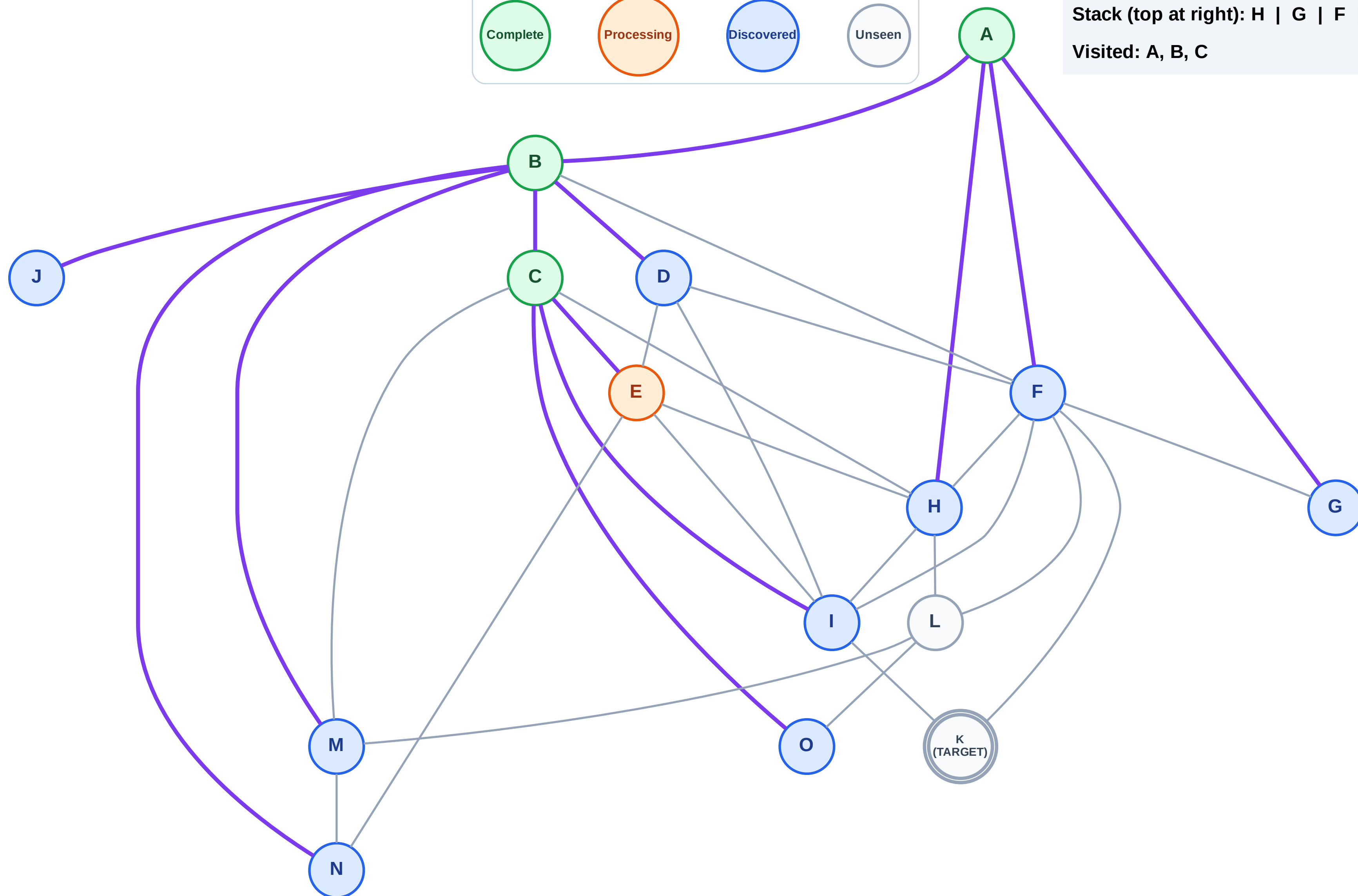
Step 8: Process node E

Step 8: Process node E



Stack (top at right): H | G | F | N | M | J | D | O | I

Visited: A, B, C



DFS Traversal

Step 9: No new neighbors from E

Legend

Complete

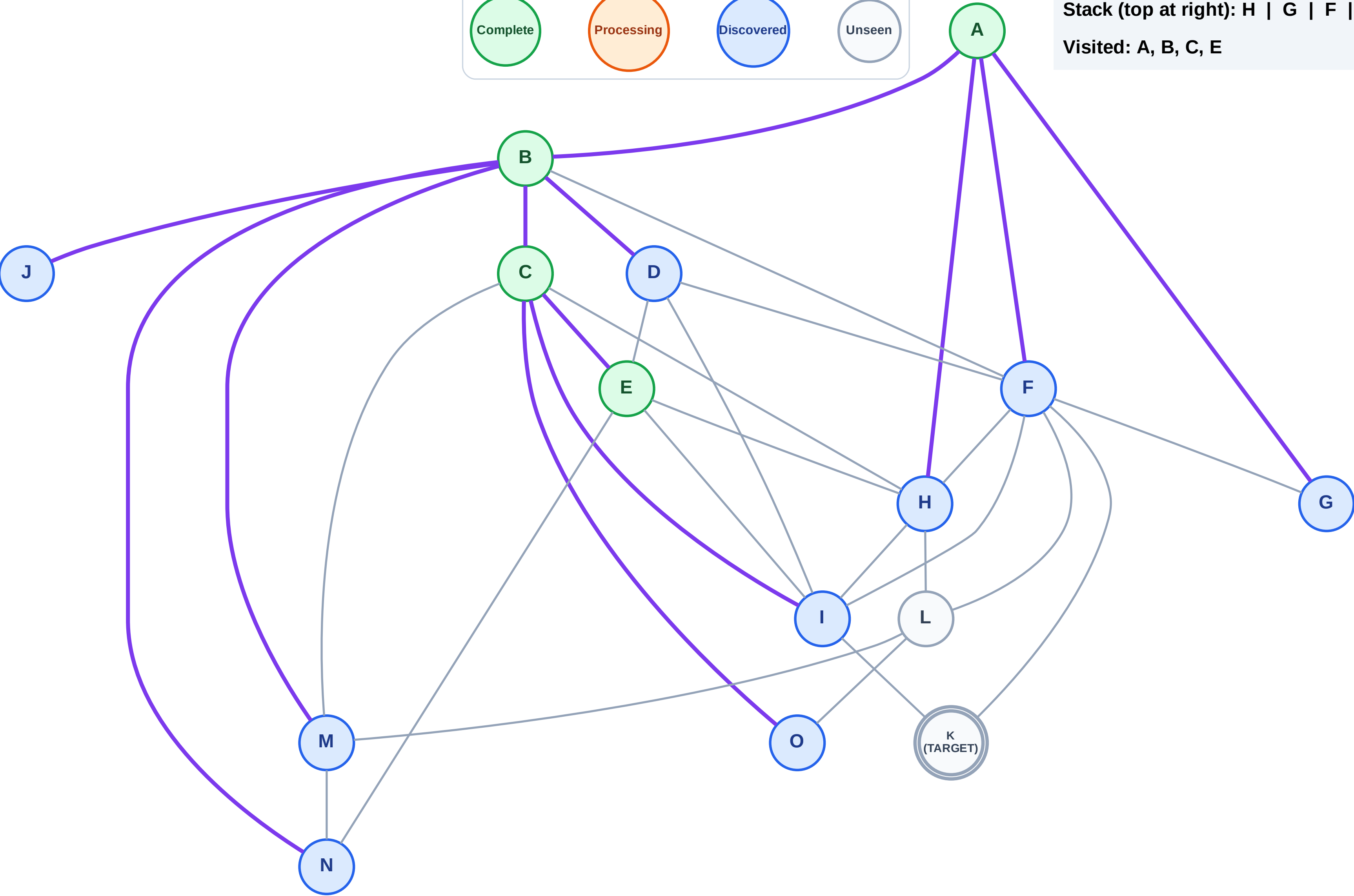
Processing

Discovered

Unseen

Stack (top at right): H | G | F | N | M | J | D | O | I

Visited: A, B, C, E



Step 10: Process node I

Step 10: Process node I

Legend

Complete

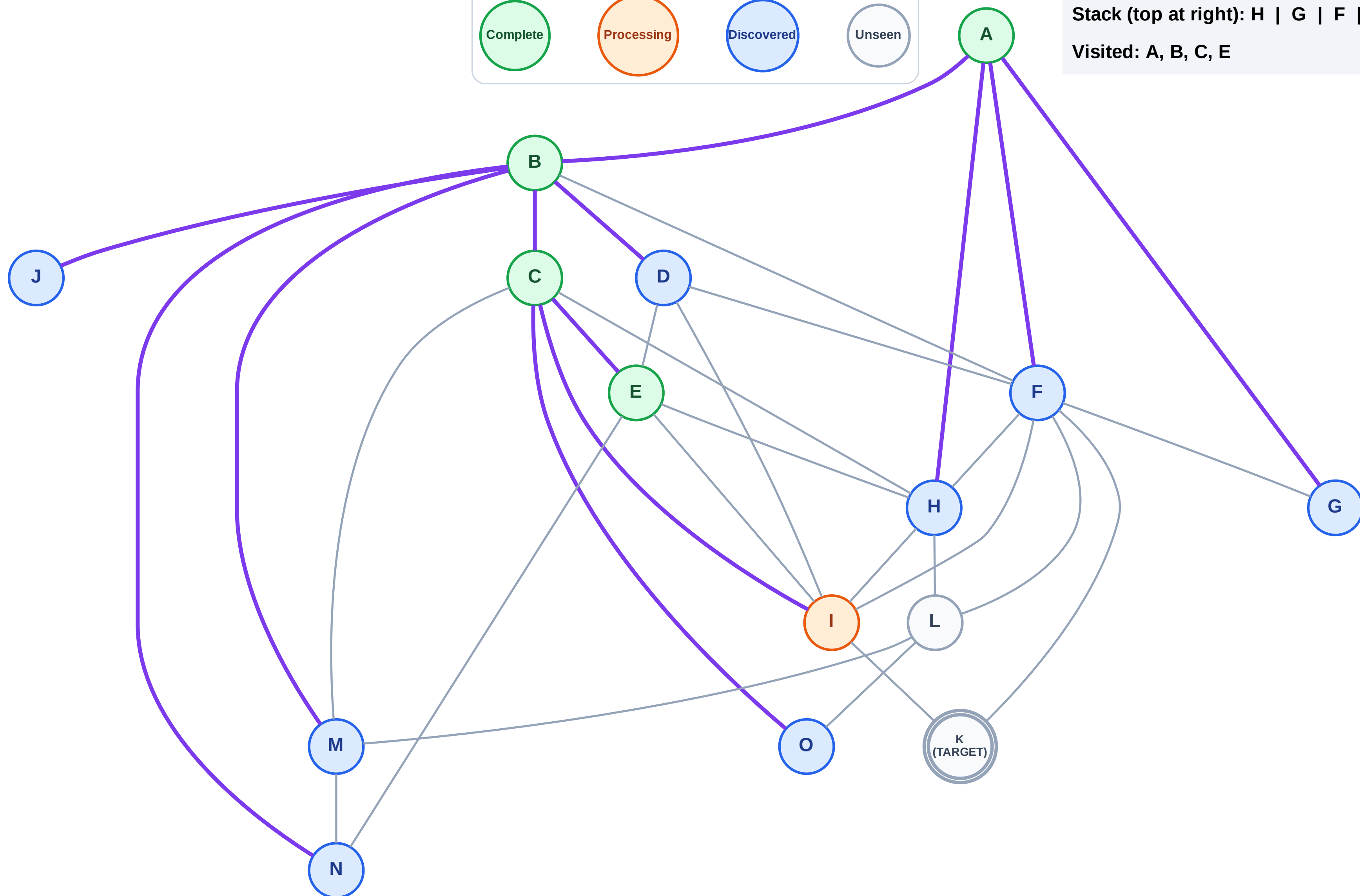
Processing

Discovered

Unseen

Stack (top at right): H | G | F | N | M | J | D | O

Visited: A, B, C, E

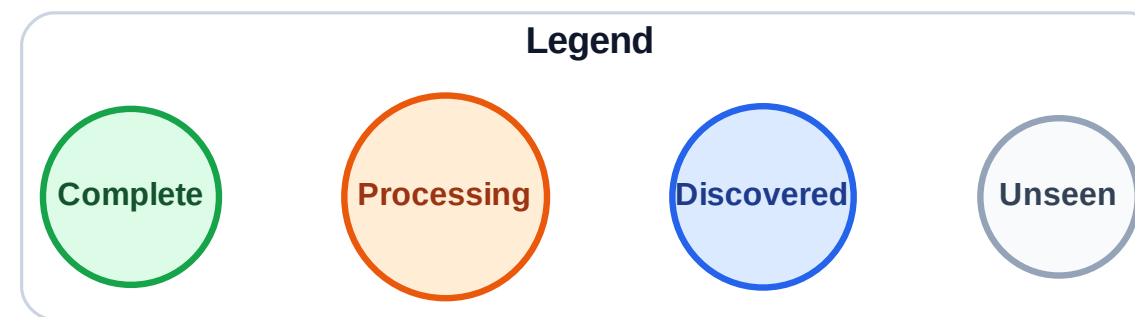


DFS Traversal

Step 11: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (4 hops): A → B → C → I → K

DFS Traversal

Step 11: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (4 hops): A → B → C → I → K



Stack (top at right): H | G | F | N | M | J | D | O

Visited: A, B, C, E, I, K

